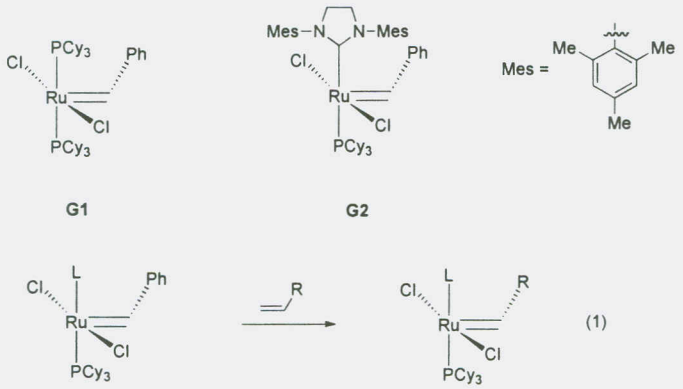
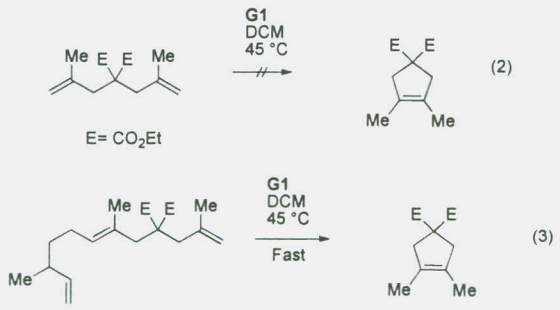
Organic Cume
3-6-09

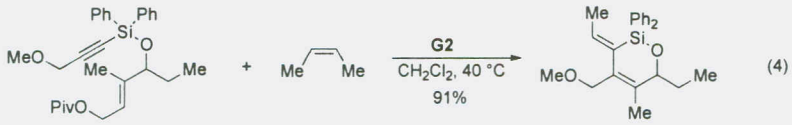
1. Robert H. Grubbs at Caltech has developed two generations of widely used Ru-based catalysts (**G1** and **G2**) for Olefin Metathesis (*J. Am. Chem. Soc.* **2001**, *123*, 6543-6554). Please provide a catalytic cycle for the metathesis reaction shown in eq 1. **G2** is known to be significantly more reactive than **G1**. Please use the catalytic cycle you propose to account for the rate enhancement of **G2**.



2. Thomas R. Hoye at the University of Minnesota developed a strategy, Relay Ring-Closing Metathesis (RRCM), to circumvent difficult Ring-Closing Metathesis reactions such as eq 2 (*J. Am. Chem. Soc.* **2004**, *126*, 10210-10211). One example from this communication paper is shown below (eq 3). Please provide a detailed mechanism for the reaction shown in eq 3 and your reasoning why the strategy worked (Note: DCM is CH_2Cl_2).



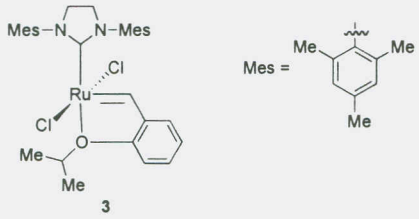
3. Daesung Lee at the University of Wisconsin, Madison, developed a tandem sequence of metathesis reactions to prepare synthetically useful silylated 1,3-dienes (*J. Am. Chem. Soc.* **2005**, *127*, 9410-9415). Please provide a plausible mechanism for the reaction shown in eq 4. Additionally, please specify the two metathesis processes involved in the sequence and comment on the role of the silyl group.



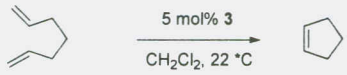
(4)

Note: *cis*-2-Butene was directly bubbled into the reaction.

4. Amir H. Hoveyda at Boston College developed a variant (**3**) of Grubbs' 2nd generation catalyst (**G2**) that is active and recyclable (*J. Am. Chem. Soc.* **2000**, *122*, 8168-8179). Please use eq 5 to propose a catalytic cycle for the ring-closing metathesis (RCM) reaction and then use your proposed catalytic cycle to rationalize why **3** is recyclable.



3



(5)